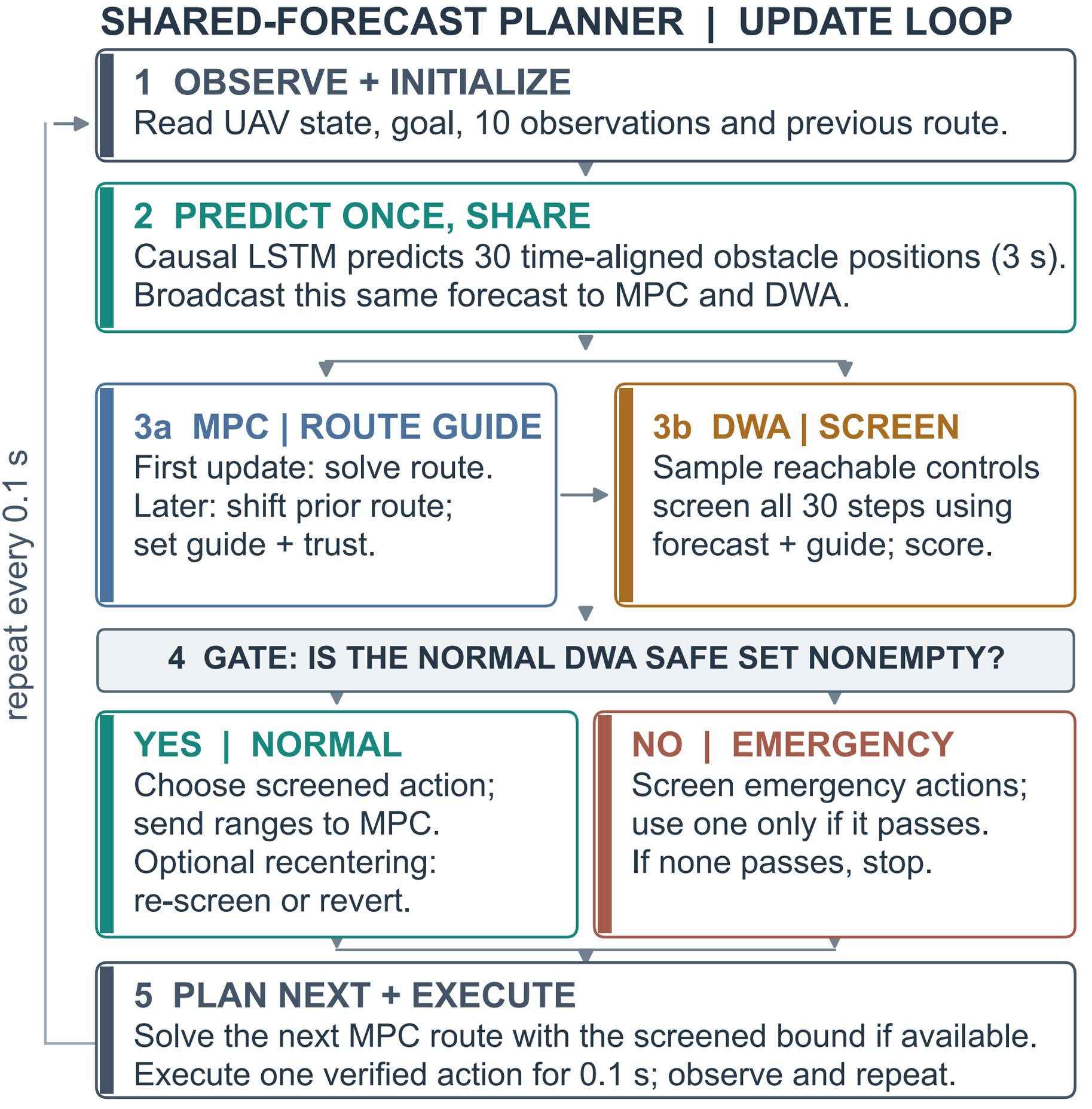
SHARED-FORECAST PLANNER | UPDATE LOOP
1 OBSERVE + INITIALIZE
Read UAV state, goal, 10 observations and previous route.
2 PREDICT ONCE, SHARE
Causal LSTM predicts 30 time-aligned obstacle positions (3 s).
Broadcast this same forecast to MPC and DWA.
3a MPC | ROUTE GUIDE
First update: solve route.
Later: shift prior route;
set guide + trust.
3b DWA | SCREEN
Sample reachable controls
screen all 30 steps using
forecast + guide; score.
4 GATE: IS THE NORMAL DWA SAFE SET NONEMPTY?
YES | NORMAL
Choose screened action;
send ranges to MPC.
Optional recentering:
re-screen or revert.
NO | EMERGENCY
Screen emergency actions;
use one only if it passes.
If none passes, stop.
5 PLAN NEXT + EXECUTE
Solve the next MPC route with the screened bound if available.
Execute one verified action for 0.1 s; observe and repeat.
repeat every 0.1 s